GCSE · OCR · Computer Science

2 hours 35 questions

Wired & Wireless Networks, Protocols & Layers

Wired & Wireless Networks, Protocols & Layers

Modes of Connection / Encryption / IP Addressing & MAC Addressing / Standards in Computing / Network Protocols / The Concept of Layers

Easy (12 questions)	/23
Medium (14 questions)	/46
Hard (9 questions)	/27
Total Marks	**/96**

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Easy Questions

1 **Protocols** are used to transmit data through the network and over the internet.

Identify one protocol that can be used to perform each of the following tasks:

- Send an email
- Access a website securely

.....

..... **(2 marks)**

2 The owners of a large bakery have a **Local Area Network** (**LAN**) with a **star topology**. They order their supplies over the Internet.

When data is transmitted from the bakery to the supplier, **network protocols** are used.

Define what is meant by a '**network protocol**'.

..... **(1 mark)**

3 Data transmitted over the network uses different protocols.

Tick one box in **each row** to identify whether the protocol is related to email, transferring files or accessing websites.

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Protocol	Email	Transferring files	Accessing websites
POP			
FTP			
SMTP			
HTTPS			

(4 marks)

4 TCP/IP is one example of a protocol.

Give the name of one appropriate protocol **for each task** in the table.

Task	Protocol for this task
Sending an email from one mail server to another	
Transmitting a file from a client to a server	
Viewing a website using a web browser	
Downloading an email to your computer	

.....

.....

.....

..... **(4 marks)**

5 The following table has **descriptions of Ethernet and WiFi**.

Tick (✓) one box in each row to identify if the description is more appropriate for Ethernet or WiFi.

Description	Ethernet	WiFi
A wired connection		
More likely to be affected by interference		
Data can be transmitted at a faster speed		
Wireless transmission		
Shorter transmission range before data is lost		

(5 marks)

6 **State** what the acronym TCP/IP stands for

..... **(1 mark)**

7 **Identify** whether TCP/IP is a single protocol or a stack of protocols

..... **(1 mark)**

8 **Identify** one difference between a **Full Mesh** and a **Partial Mesh** topology

..... **(1 mark)**

9 **State** which address is **permanent** and cannot be changed on a device's Network Interface Controller (NIC)

..... **(1 mark)**

10 **Identify** the number based used for a **MAC address**

..... **(1 mark)**

11 **Define** the term 'layer' as used in computer networking

..... **(1 mark)**

12 **Identify** one benefit of using a layered approach when teaching networking

..... **(1 mark)**

Medium Questions

1 Users can access websites from library computers.

Complete the description of accessing websites using the given list of terms.

Not all terms will be used.

'0' ' 1' '127' '128' '255' ' 256' ' Colon'

'Domain Name Server' 'Embedded systems' 'File server'

'Full stop' 'Hyphen' 'Internet protocol' 'MAC address'

'Router' 'Uniform Resource Locator' 'Web server' 'Clients'

A website is hosted on a ..... The computers that access the websites are called ..... .

The user enters the ..... into a web browser. The web browser sends a request to the ..... for the matching IP (Internet Protocol) address. If found the IP address is returned. A request is then sent to this IP address.

An IPv4 address is made of 4 groups of digits. Each group can be between the denary values ..... and ..... . The groups of digits are separated by a ..... .

(7 marks)

2 A library does not use encryption when data is transmitted through the network.

Give two reasons why the library **should use encryption**.

.....

..... **(2 marks)**

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3 (a) The network has one **wireless access point** in the kitchen that transmits data on the **5GHz frequency**.

When the laptop is in the kitchen, it has better network performance.

Explain why the laptop's **network performance** is **lower in the bedroom.**

.....

..... **(2 marks)**

(b) **State two** ways Hope could **improve the wireless network performance** in the bedroom.

1	
2	

.....

..... **(2 marks)**

4 **Ethernet cables** are used within the office building.

Tick one box in each row to identify if the statement about Ethernet is True or False.

Statement	True	False
Ethernet is a protocol		
Ethernet uses wireless data transmission		
Ethernet can transmit data at speeds of up to 100Gbits per second		
Ethernet is a protocol within the TCP/IP stack		

(4 marks)

5 Computers access the Internet using the **TCP/IP model**.

The TCP/IP model **uses layers** including the application layer and transport layer.

Explain why the TCP/IP model **uses layers**.

.....

..... **(2 marks)**

6 (a) For the scenario below, **identify** the **most appropriate protocol** to be used and **explain the function** of the protocol.

A user wants to transfer a file directly from his computer to his friend's computer.

.....

..... **(2 marks)**

(b) For the scenario below, **identify** the **most appropriate protocol** to be used and **explain the function** of the protocol.

A customer wants to securely log into her bank's website to check her account balance.

.....

..... **(2 marks)**

7 (a) An airport has computers that are connected together on a Local Area Network (LAN).

Each computer has an IP address and a MAC address.

Give **one valid** example of an IPv4 address and **one valid** example of an IPv6 address.

.....

..... **(2 marks)**

(b) **Describe** the format of a MAC address.

.....

..... **(2 marks)**

8 (a) The airport currently has wired connections in their Local Area Network.

Describe **two** benefits to the airport of using wired connections in their network.

.....

.....

.....

..... **(4 marks)**

(b) **Explain** the reasons why the airport should also allow the network to be accessed using a wireless connection.

.....

.....

..... **(3 marks)**

9 **Describe** the purpose of the TCP/IP model in a network

.....

..... **(2 marks)**

10 **Explain** why the TCP/IP model uses **layers**

.....

..... **(2 marks)**

11 **Explain** one advantage of using a Mesh topology in a military communication system

.....

..... **(2 marks)**

12 **Describe** one disadvantage of implementing a **Full Mesh** network in a large office

.....

..... **(2 marks)**

13 **Explain** the difference in the **purpose** of an IP address versus a MAC address

.....

..... **(2 marks)**

14 **Describe** how layers promote **interoperability** between different hardware vendors

.....

..... **(2 marks)**

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Hard Questions

1 The wired connection is an **Ethernet connection**.

Ethernet is **considered a standard**.

Explain why Ethernet is a standard.

.....

..... **(2 marks)**

2 (a) TCP/IP is a set of protocols **based on layers**.

With regards to network protocols, **define** what is meant by a '**layer**'.

..... **(1 mark)**

(b) **Describe one advantage** of **using layers** to construct network protocols.

.....

..... **(2 marks)**

3 **Explain** the difference between how the **IMAP** (Internet message access protocol) and **SMTP** (simple mail transfer protocol) **protocols are used**.

.....

..... **(2 marks)**

4 **"A file is sent from an office in London to a client in New York."**

Analyse the role of the TCP/IP stack in ensuring the data arrives reliably

.....

.....

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.....

..... **(4 marks)**

5 **Evaluate** the benefit of the TCP/IP model being a **standard** rather than a proprietary system.

.....

.....

..... **(3 marks)**

6 **Compare** the performance of a Mesh topology and a Star topology when the **central switch** fails

.....

.....

.....

..... **(4 marks)**

7 A city-wide emergency sensor network uses a wireless Mesh.

Justify this choice over a Star topology

.....

.....

..... **(3 marks)**

8 **"A router receives a data packet from the internet."**

Describe how the router uses both IP and MAC addresses to deliver the packet to the correct laptop in a house

.....

.....

.....**(3 marks)**

9 "A network administrator is configuring a new laptop to connect to the company network."

Describe the format of a MAC address and how it is used to identify devices on a local network.

.....

.....

.....**(3 marks)**